Cyber crime

CASE STUDIES

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CASE 1 — WANNACRY RANSOMWARE ATTACK (GLOBAL)



Illustrative graphic of a ransomware screen. Replace with the original newspaper clipping if your teacher requires one.

IMAGE DESCRIPTION

The graphic shows the kind of message victims saw during the WannaCry outbreak: files locked, a payment demand in Bitcoin and a countdown timer. It represents one of the first ransomware attacks to spread across the world automatically, without any user needing to click a link.

1) NEWS HISTORY — WHAT HAPPENED?

On 12 May 2017 the WannaCry ransomware began spreading across the internet, encrypting files on Windows computers and demanding a Bitcoin ransom (commonly reported as US$300, rising later). It affected more than 200,000 computers in roughly 150 countries. In the United Kingdom, parts of the National Health Service (NHS) were disrupted, and appointments and operations were cancelled. Companies in logistics, telecom and manufacturing were also hit. A security researcher found a hard-coded web address in the malware that acted as a "kill switch"; registering it slowed the spread within days. UK and US governments later attributed the attack to groups linked to North Korea.

SOURCE / NEWS REFERENCE

UK National Audit Office, "Investigation: WannaCry cyber attack and the NHS" (2018); Microsoft Security Response Center guidance on bulletin MS17-010 (2017). The image above is an illustrative recreation, not a photograph of the original screen.

2) ATTACK TECHNIQUE

WannaCry combined ransomware with a worm. It spread through the Server Message Block (SMBv1) vulnerability using an exploit known as EternalBlue, for which Microsoft had already released a patch in March 2017. Unpatched and unsupported systems were therefore the easiest targets. Once inside a machine, the malware encrypted files and then scanned for other vulnerable computers on the network and the internet.

3) TOTAL LOSS / FINANCIAL IMPACT

Exact global losses are uncertain and estimates vary widely; some industry reports mention figures in the billions of US dollars. The NHS later reported costs of about £92 million, including lost activity and IT recovery. The attack also caused cancelled treatments, stopped production lines and heavy loss of working time. The ransom payments themselves were comparatively small.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Organisations isolated infected machines, restored systems from clean backups and applied the missing patch. Microsoft took the unusual step of releasing emergency updates even for older unsupported Windows versions. Researchers analysed the malware, and the kill-switch domain limited further spread. Law-enforcement agencies and national cyber centres coordinated the investigation.

5) RECOVERY COST AND RECOVERY TIME

Recovery took days for many organisations and longer for hospitals that had to rebuild systems. The NHS figure above is one documented estimate; a single worldwide recovery cost has not been confirmed. Costs included forensic work, replacement hardware, overtime, patching programmes and delayed patient care.

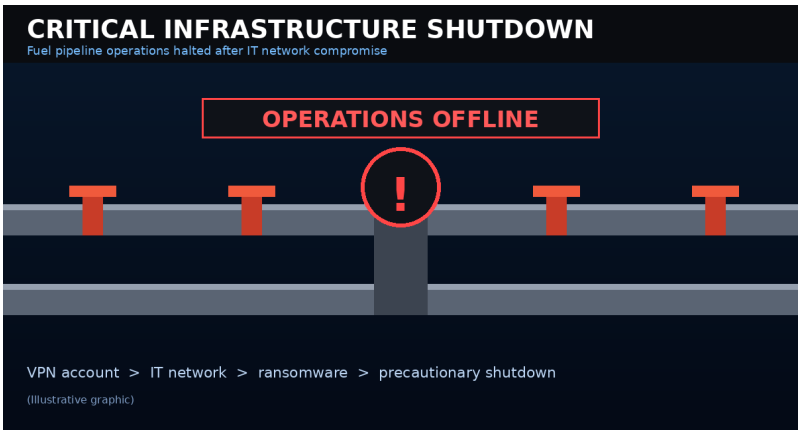
6) HOW COULD IT HAVE BEEN PREVENTED?

Apply security updates promptly; disable SMBv1 and block unnecessary SMB traffic from the internet; keep offline and tested backups; segment networks; replace unsupported operating systems; use endpoint detection tools; and rehearse an incident response plan.

SOLUTION / KEY LESSON

A patch that exists but is not installed gives no protection. Patch management, network segmentation and offline backups are basic controls that can stop a worm-like attack from becoming a national emergency.

CASE 2 — COLONIAL PIPELINE RANSOMWARE ATTACK (USA)



Illustrative graphic showing a fuel pipeline shutdown after an IT network compromise.

IMAGE DESCRIPTION

The graphic represents a critical-infrastructure operator that shut down its pipeline operations as a precaution after ransomware reached its business systems. It highlights that cyber attacks on IT networks can affect everyday physical services such as fuel supply.

1) NEWS HISTORY — WHAT HAPPENED?

In May 2021 Colonial Pipeline, a major US fuel pipeline operator, found ransomware on its computer systems and halted pipeline operations as a precaution. The attack was attributed to a criminal group called DarkSide. Fuel shortages, panic buying and higher prices were reported in several southeastern US states. Pipeline operations restarted after about six days. The company also paid a ransom, which later drew intense public and political debate.

SOURCE / NEWS REFERENCE

U.S. Department of Justice press release on the recovery of Colonial Pipeline ransom funds (June 2021); U.S. Cybersecurity and Infrastructure Security Agency (CISA) advisory on DarkSide ransomware (2021).

2) ATTACK TECHNIQUE

Public reporting indicates the attackers entered through a compromised Virtual Private Network (VPN) account protected only by a password, with no multi-factor authentication, and the account was reportedly no longer in active use. The attackers then moved through the network, reportedly stole data and deployed ransomware in a double-extortion style. The exact origin of the leaked password has not been publicly confirmed.

3) TOTAL LOSS / FINANCIAL IMPACT

Colonial reportedly paid about 75 Bitcoin (roughly US$4.4 million at the time). The larger loss was economic and social: interrupted fuel deliveries, higher prices and emergency measures by authorities. The US Department of Justice later announced it had recovered a large part of the ransom, about 63.7 Bitcoin.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

The company shut down pipeline operations, hired outside cyber-security experts and informed federal authorities. The US government issued emergency fuel-transport waivers, and agencies investigated the group behind the attack. Operations were gradually restored, and the company later strengthened its security programme.

5) RECOVERY COST AND RECOVERY TIME

The ransom payment was reported at about US$4.4 million, but a complete recovery cost is not publicly confirmed. Pipeline operations resumed after about six days, while the return of normal fuel supply took longer. Additional costs include investigation, system rebuilding, security upgrades and legal exposure.

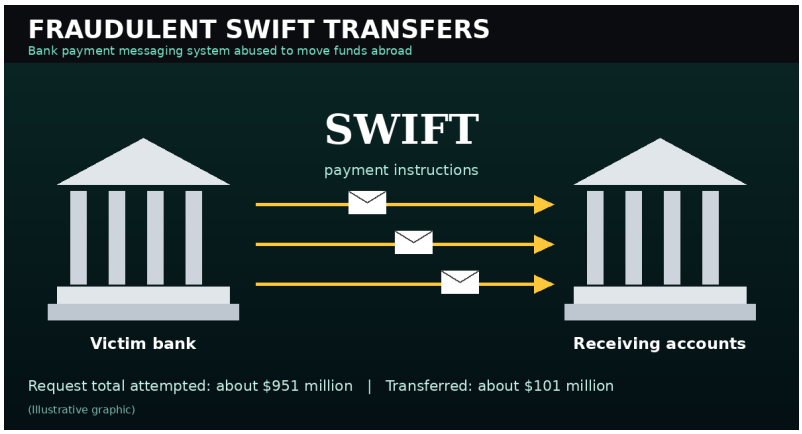
6) HOW COULD IT HAVE BEEN PREVENTED?

Require multi-factor authentication on all remote access; remove unused accounts quickly; use unique strong passwords; separate IT and operational networks; monitor remote logins; test backups; and maintain a tested incident response plan for critical operations.

SOLUTION / KEY LESSON

One weak remote-access account can disrupt a nation’s critical infrastructure. Multi-factor authentication and account clean-up are simple, low-cost defences, and operators of critical services must plan how to keep running when IT systems fail.

CASE 3 — BANGLADESH BANK SWIFT HEIST (BANGLADESH / PHILIPPINES)



Illustrative graphic showing fraudulent payment instructions sent through a bank messaging system.

IMAGE DESCRIPTION

The graphic illustrates how attackers used the SWIFT interbank messaging network to send fraudulent payment instructions from a central bank's account to accounts abroad. It shows that attackers target the business processes around money, not only customer data.

1) NEWS HISTORY — WHAT HAPPENED?

In February 2016, attackers sent about three dozen fraudulent payment requests, worth roughly US$951 million, from the Bangladesh Bank account at the Federal Reserve Bank of New York. Several requests were blocked, partly because a bank noticed a spelling mistake in one instruction and asked questions. About US$101 million was still transferred; around US$20 million sent to Sri Lanka was recovered, while roughly US$81 million sent to the Philippines was largely laundered through casinos and was only partly recovered.

SOURCE / NEWS REFERENCE

Reuters investigative reporting on the Bangladesh Bank heist (2016); BAE Systems and SWIFT security advisories on the attack (2016). Figures are rounded and reported in public sources.

2) ATTACK TECHNIQUE

The attackers first gained access to the bank's internal network, studied how payments were made and obtained credentials used to authorise SWIFT messages. Malware was reportedly used to interfere with the bank's payment-confirmation printouts so that staff did not notice the transfers quickly. The attack was timed over a weekend and public holiday period when staff were away in different time zones.

3) TOTAL LOSS / FINANCIAL IMPACT

Roughly US$81 million was lost after reaching the Philippines, making it one of the largest cyber-enabled bank thefts reported. Only a small part of that money was later recovered. The incident also damaged confidence in the bank, led to resignations and caused banks worldwide to review payment-system security.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Bangladesh Bank asked the Federal Reserve Bank of New York and other banks to halt transfers, and requested help from authorities in several countries. Investigations took place in Bangladesh, the Philippines and the US. SWIFT introduced stronger security requirements for its member banks after similar incidents.

5) RECOVERY COST AND RECOVERY TIME

A single complete recovery cost is not publicly confirmed. The direct theft was about US$81 million after partial recoveries, and the investigation, forensic work, legal action and new security controls continued for years. Money laundering through casinos made tracing and recovery difficult.

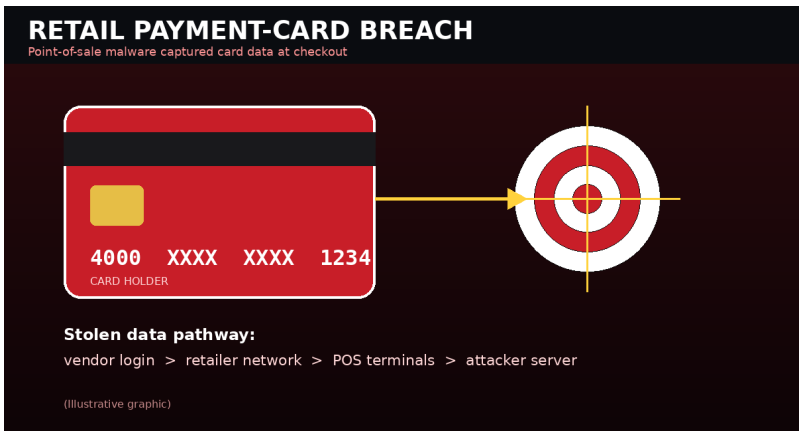
6) HOW COULD IT HAVE BEEN PREVENTED?

Isolate the payment-messaging environment; require multi-person approval and multi-factor authentication for payment release; monitor transactions for unusual patterns; keep staff and monitoring active during holidays; keep logs and printouts tamper-resistant; and run regular security assessments and incident drills.

SOLUTION / KEY LESSON

Financial systems need layered controls around people, processes and technology. Even a strong payment network is unsafe if the connected systems and approval procedures at a bank are weak.

CASE 4 — TARGET RETAIL DATA BREACH (USA)



Illustrative graphic of payment-card data being stolen through a retailer's point-of-sale network.

IMAGE DESCRIPTION

The graphic represents a payment-card breach at a large retailer. Attackers used a third-party vendor's access to enter the network and then captured card details from checkout systems. It is a well-known example of supply-chain risk.

1) NEWS HISTORY — WHAT HAPPENED?

Between late November and mid-December 2013, attackers stole payment-card data from Target, a major US retailer, during the busy holiday shopping season. Target announced that about 40 million credit and debit card records were affected, and later said that personal information, such as names and contact details, for about 70 million customers was also taken. The breach became widely discussed in business and security circles and led to senior leadership changes.

SOURCE / NEWS REFERENCE

U.S. Senate Committee on Commerce, Science and Transportation, "A 'Kill Chain' Analysis of the 2013 Target Data Breach" (2014); Target company announcements (2013–2014).

2) ATTACK TECHNIQUE

The attackers reportedly stole network credentials from a third-party heating-and-cooling vendor that had access to Target's systems. They moved from a vendor portal into the internal network and installed malware on point-of-sale (POS) terminals. The malware captured card data from memory while cards were being processed, and the stolen data was collected on internal servers before being sent out.

3) TOTAL LOSS / FINANCIAL IMPACT

Beyond fraud losses to banks and cardholders, Target faced investigations, lawsuits and a drop in customer trust. Target later agreed to an US$18.5 million settlement with a group of US states (2017). The company also reported very large breach-related expenses over the following years, though the exact total depends on how insurance recoveries are counted.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Target removed the malware, notified customers and offered free credit monitoring. It invested in new security technology and staff, and moved toward chip-and-PIN payment cards. The company's chief executive resigned in 2014, and regulators and states pursued legal action.

5) RECOVERY COST AND RECOVERY TIME

A single recovery figure is not confirmed. Reported costs include the US$18.5 million multistate settlement, legal fees, credit-monitoring services, card re-issue costs borne by banks and the long-term investment in new security controls. Recovery of customer trust took considerably longer than technical clean-up.

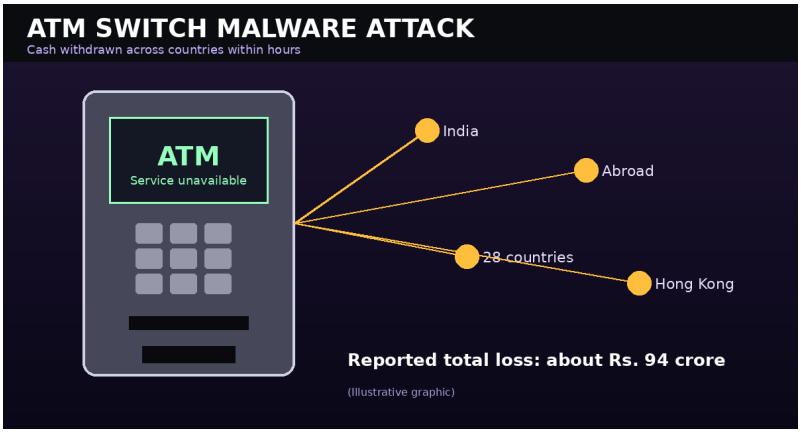
6) HOW COULD IT HAVE BEEN PREVENTED?

Limit and monitor vendor access; require multi-factor authentication for third parties; segment payment systems from the rest of the network; whitelist approved software on POS terminals; act on security-tool alerts; and use point-to-point encryption and chip-based cards.

SOLUTION / KEY LESSON

An organisation is only as secure as its weakest partner. Third-party access must be restricted, monitored and treated as part of the company’s own security perimeter, and alerts from security tools must be investigated.

CASE 5 — COSMOS BANK MALWARE ATTACK (PUNE, INDIA)



Illustrative graphic of an ATM-switch malware attack with withdrawals across many locations.

IMAGE DESCRIPTION

The graphic represents a cyber attack on an Indian co-operative bank's ATM payment system. Within a short period, cash was withdrawn using cloned or fake card transactions at ATMs in India and abroad. It shows how a compromise of backend payment servers can cause large losses quickly.

1) NEWS HISTORY — WHAT HAPPENED?

In August 2018, Cosmos Co-operative Bank in Pune reported a cyber attack in which about Rs. 94 crore was stolen over a weekend. Reports said that fraudulent card transactions were carried out across around 28 countries as well as in India, and a further amount was moved through unauthorised SWIFT transfers to an overseas account. The incident was investigated by the Pune police and covered widely by Indian media.

SOURCE / NEWS REFERENCE

Indian press coverage (Times of India, Indian Express, August 2018) and statements from the Pune police cyber cell. Amounts are rounded and as reported in public sources; check the latest official records before quoting exact numbers.

2) ATTACK TECHNIQUE

Reports said that malware was introduced into the bank's ATM switch server, allowing attackers to approve fraudulent withdrawal requests for cloned debit cards without genuine authorisation checks. The same attack reportedly extended to the bank's SWIFT messaging system. The exact entry point was not fully described in public reports, so it is not claimed here.

3) TOTAL LOSS / FINANCIAL IMPACT

About Rs. 94 crore was reportedly stolen in total, mainly through thousands of card transactions at ATMs in India and overseas, plus unauthorised SWIFT transfers. The bank's customers faced temporary service disruption, and public confidence in smaller co-operative banks was questioned.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

The bank shut down its ATM and card services, informed the Reserve Bank of India and police, and began a forensic investigation with security experts. Authorities worked to trace and block some of the funds. Customers' account balances were reported to be protected by the bank, and services were restored in stages.

5) RECOVERY COST AND RECOVERY TIME

The precise recovery cost and the amount finally recovered are not confirmed in the public sources used here, and only part of the stolen money was reported as recovered. Costs included forensic analysis, system rebuilding, customer support and regulatory compliance work. Service restoration took days to weeks.

6) HOW COULD IT HAVE BEEN PREVENTED?

Segment and harden ATM switch and SWIFT servers; apply security patches; require strong authentication and multi-person approval for payments; monitor card transactions for abnormal patterns and geography; limit international withdrawals; use chip-based cards; and run regular security audits.

SOLUTION / KEY LESSON

Banks of every size must secure their core payment infrastructure. Real-time monitoring of card and payment activity can detect unusual withdrawals early enough to limit losses.